

DEV PANKAJBHAI GOTI

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SUMMARY

Robotics engineer specializing in simulation-to-hardware validation for autonomous systems, building multi-sensor perception and navigation pipelines with research spanning SLAM, sensor fusion, and reinforcement learning

EDUCATION

Texas A&M University

College Station, TX

Master of Science in Electrical Engineering | GPA : 4.0/4.0

May 2027

Relevant Coursework: Visual Navigation, Computer Vision, Robotics & Spatial Intelligence, Reinforcement Learning

National Institute of Technology Karnataka, Surathkal

Mangalore, India

Bachelor of Technology in Electrical and Electronics Engineering | GPA : 3.71/4.0

May 2025

EXPERIENCE

Piaggio Fast Forward

Boston, MA

Robotics and AI Co-op

Jul 2026 – Dec 2026

- Engineered an end-to-end SIL simulation stack connecting production autonomy and motion-control software with ROS2 Jazzy and NVIDIA Isaac Sim.
- Integrated RGB/depth cameras, LiDAR, IMU, odometry, and robot control through ROS2 Action Graphs and reusable USD assets to validate perception, tracking, and autonomous navigation workflows in simulation
- Profiled and optimized Isaac Sim across PhysX, RTX rendering, CPU/GPU workloads, sensor update rates, collision geometry, and timing synchronization to improve runtime efficiency and support high-frequency robotics pipelines
- Debugged and integrated odometry, sensor management, and simulation tooling, including fixes for simulation-specific timestamp handling and trajectory drift across the autonomy stack

Indian Institute of Space Science and Technology

Thiruvananthapuram, India

Summer Research Intern – IASc Summer Research Fellowship Programme

May 2024 – Jul 2024

- Developed a PPO-based DRL trajectory planner for a 7-DOF robotic arm enabling collision-aware and smooth goal-directed motion in continuous state-action spaces without explicit motion planning models
- Evaluated learned policies in simulation using reward convergence, target accuracy, and trajectory smoothness, demonstrating reliable control performance for high-DOF robotic manipulation

TECHNICAL SKILLS

Programming: Python (Advanced), C (Intermediate), C++ (Advanced)

Tools: Isaac Sim, ROS2, MATLAB, Simulink, Gazebo, RViz, VS Code, Conda, Jupyter, Docker, Linux, Git, GitHub, CMake, Colcon, Nav2, CUDA, TensorRT

Libraries: NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn, TensorFlow, PyTorch, Keras, Open3D, PCL

Robotics: Motion Planning, SLAM, Localization, Multimodal Sensor Fusion, State Estimation, Kalman Filter

PROJECTS

Robust Outlier Rejection in Visual SLAM using GNC-TLS

- Implemented GNC-TLS optimization for loop closure outlier rejection in a Visual SLAM pipeline, achieving stable absolute trajectory error (ATE) of 4.75–7.37m under 36% artificially injected KITTI outliers
- Benchmarked against RANSAC and chi-squared rejection baselines across multiple KITTI sequences, demonstrating ~94% trajectory error reduction and ~290× faster runtime performance

LiDAR-Camera Fusion for 3D Obstacle Detection

- Built a LiDAR-camera sensor fusion pipeline using RANSAC plane fitting, voxel grid downsampling, and DBSCAN clustering for accurate 3D obstacle detection and scene understanding
- Performed extrinsic calibration and evaluated detections using IoU-based metrics for validation and tuning

Control Barrier Functions for Safe RL Quadrotor Control

- Designed motor-level Control Barrier Function (CBF) safety filters integrated with RL-based quadrotor controllers to enforce constraint satisfaction under distribution shifts and dynamic uncertainty
- Reduced safety violations by up to 2.4× across PPO and adaptive policies in high-fidelity drone simulations

Spatially-Aware Image Editing via Multi-Model Pipeline

- Developed a VLM-orchestrated multi-model pipeline combining DINO and SAM for open-vocabulary object grounding with geometry-consistent inpainting for spatially accurate scene edits
- Achieved 99.3% intent accuracy and high-fidelity edits (SSIM: 0.976) using hybrid LaMa + SDXL blending